

Final Project: Day 120

Volatility-Aware Trading Strategy with Option Overlay

1 Introduction

This project integrates signal generation, volatility modeling, and derivatives overlay into a unified trading framework.

The goal is to build a robust strategy that adapts to market conditions and manages risk dynamically.

2 Framework Overview

1. Signal Generation
2. Volatility Forecasting
3. Position Sizing
4. Option Overlay
5. Risk Management

3 Signal Model

$$\hat{r}_{t+1} = f(X_t)$$

$$\text{Signal}_t = \begin{cases} +1 & \text{if } \hat{r}_{t+1} > 0 \\ -1 & \text{if } \hat{r}_{t+1} < 0 \end{cases}$$

4 Volatility Model (GARCH)

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

5 Position Sizing

$$w_t = \frac{\text{Signal}_t}{\sigma_t}$$

6 Option Overlay

We introduce a volatility hedge using options:

$$\Pi = w_t S_t + \theta C_t$$

where C_t is the option price.

7 Risk Objective

- Reduce downside risk
- Capture volatility spikes

8 Portfolio Return

$$R_t = w_t r_t + \theta \Delta C_t$$

9 Key Insights

- Signals drive direction
- Volatility controls exposure
- Options provide convexity

10 Conclusion

Combining statistical models with derivatives leads to more robust and adaptive trading strategies.